

# Rendering Optimization

## Problem

- One Mesh per voxel → one draw call per voxel
- Draw calls create CPU bottlenecks

## Solution

- Group voxels by color
- Render with InstancedMesh
- One draw call renders many cubes

## Trade-off

- Raycasting returns instancedId
- Need mapping: instancedId → voxel key

## VoxelStore

- Voxel, (coordinate, colorId)
- Map<CoordKey, Voxel>
  - Fast lookup and updates

## Render pipeline

VoxelStore



Voxel[]



GroupRender(colorId, voxels, indexToKey)



InstancedMesh *with JSX*

# Undo / Redo

## Data layer

### VoxelStatus

- coordinates[]
- colorId

### Operation

- type
- timestamp
- prevStatus
- newStatus

## Execution

### New operation

- push to undo stack
- clear redo stack

### Undo

- pop undo
- apply prevStatus
- push to redo

### Redo

- pop redo
- apply newStatus
- push to undo

### Load scene

- clear both stacks

### DSL

- one operation can modify multiple voxels.

